Introduction to Julia for High-Performance Computing

Carsten Bauer @ KIT, Karlsruhe

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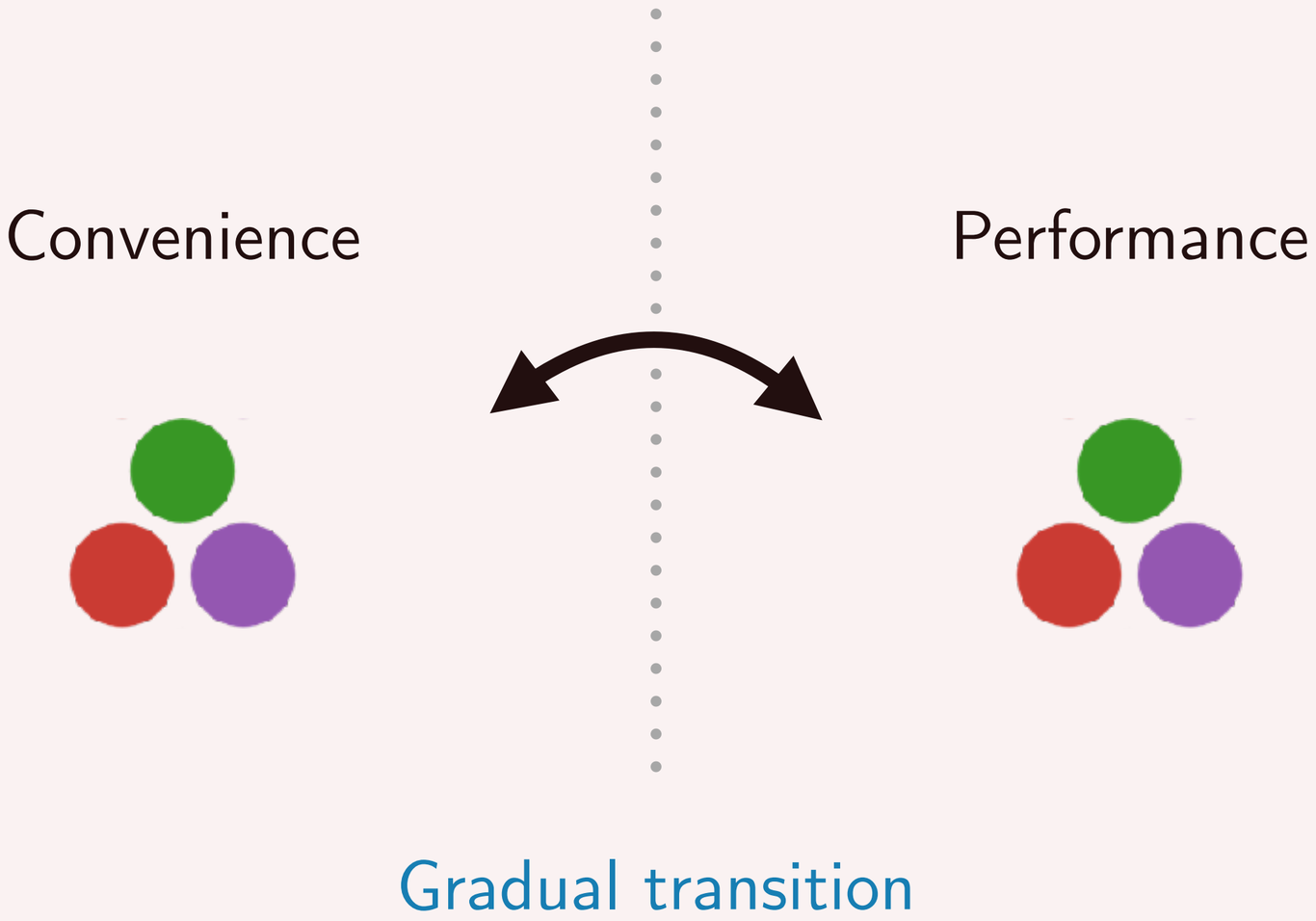
Convenience

Performance

Language Barrier


Convenience
Performance
Gradual transition

	Tuesday	Wednesday	Thursday	Friday
	Foundations	**Core**	**Node**	**Cluster**
09:00 - 10:45	Intro Onboarding	Type & Memory Optimizations	Multithreading	Distributed Computing
10:45 - 11:00	Break	Break	Break	Break
11:00 - 12:30	Fundamentals	Exercises	Exercises	Exercises
12:30 - 14:00	Lunch	Lunch	Lunch	Lunch
14:00 - 15:30	Compilation	SIMD	GPU Computing	Profiling
15:30 - 15:45	Break	Break	Break	Outro
15:45 - 17:00	Exercises	Exercises	Exercises	

Quick Live Survey

Julia’s Weaknesses

HPC with Julia is currently a niche.

Join us at JuliaCon ...

... or other conferences, like PASC, FOSDEM, or SC.

No great way to produce (small) binaries.

Achieving
high performance
can be tricky.

Julia's Strengths

Julia is interactive and convenient.

Julia has a great package manager

Laptop

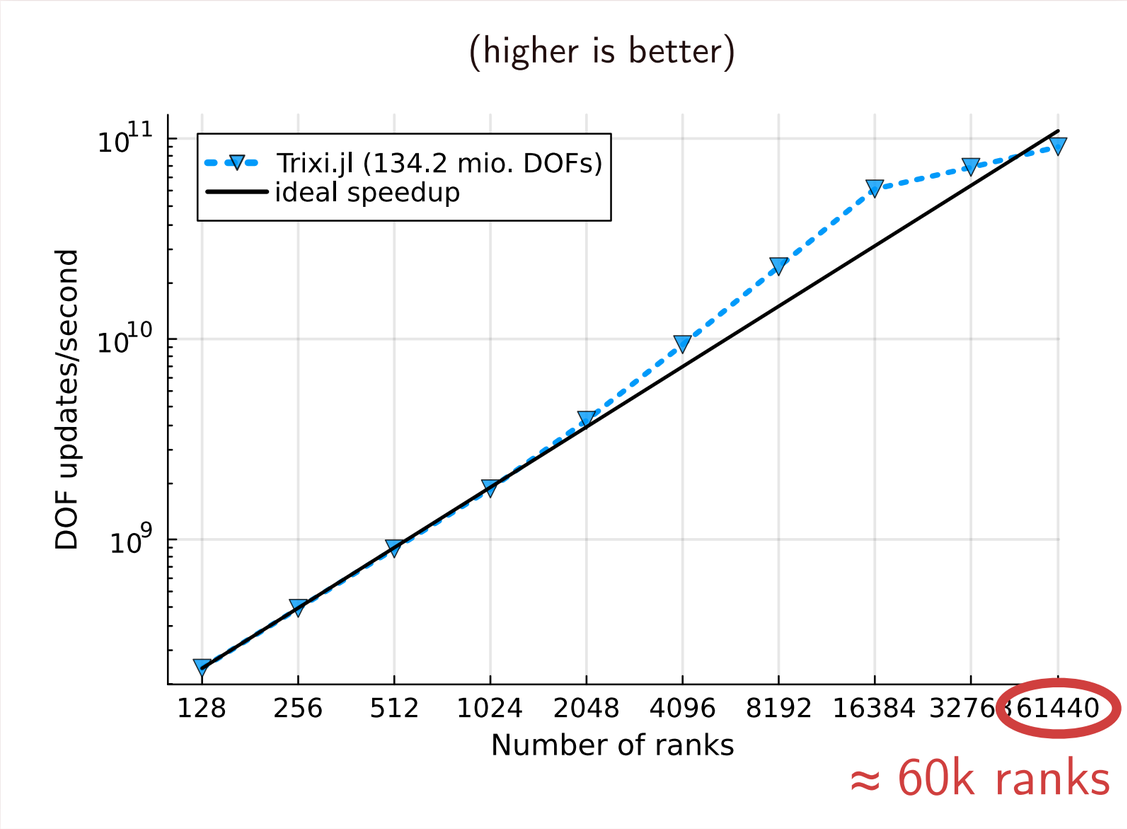
```
→  ~/myproject  tree
.
├── Manifest.toml
├── Project.toml
└── code.jl

0 directories, 3 files

→  ~/myproject  cat Project.toml
[deps]
CUDA = "052768ef-5323-5732-b1bb-66c8b64840ba"
DifferentialEquations = "0c46a032-eb83-5123-abaf-570d42b7fbaa"
MKL = "33e6dc65-8f57-5167-99aa-e5a354878fb2"
MPI = "da04e1cc-30fd-572f-bb4f-1f8673147195"

→  ~/myproject
```

HPC Cluster

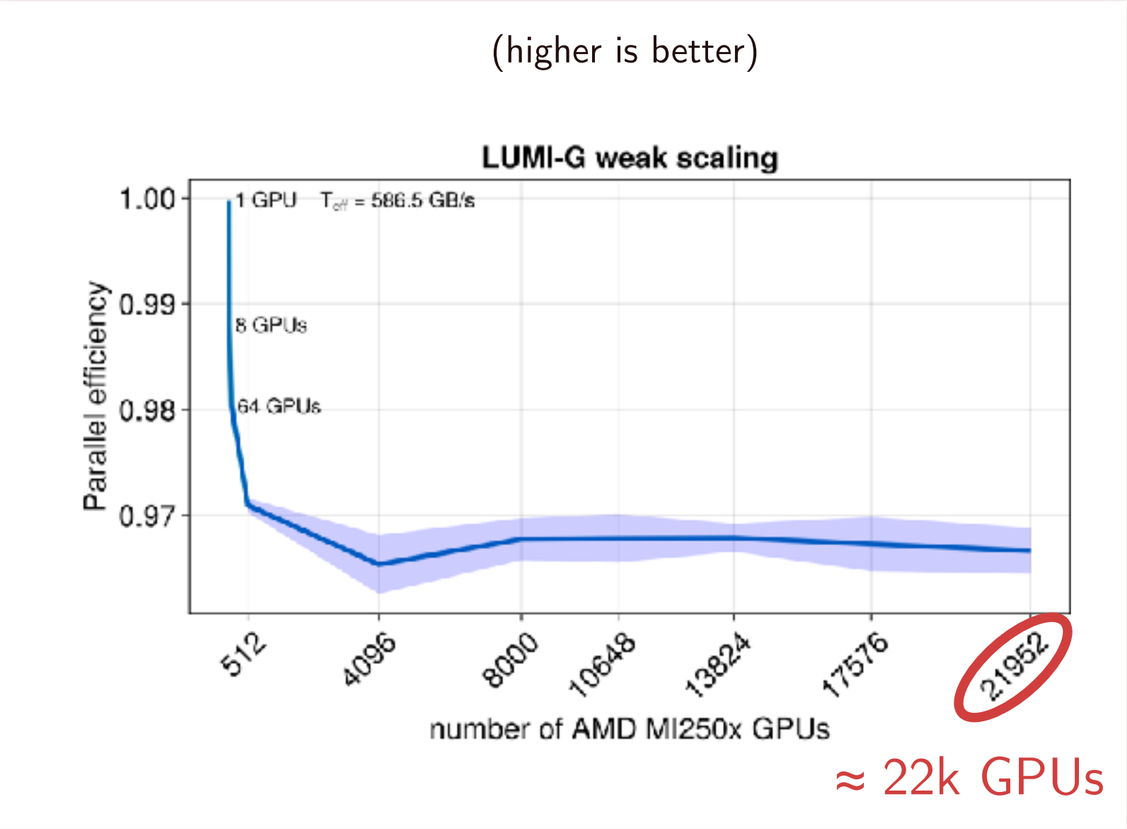
```
→  bauerc@n2login3 myproject  julia --project
               _
   _       _ _(_)_     |  Documentation: https://docs.julialang.org
  (_)     | (_) (_)    |
   _ _   _| |_  __ _   |  Type "?" for help, "]?" for Pkg help.
  | | | | | | |/ _` |  |
  | | |_| | | | (_| |  |  Version 1.7.2 (2022-02-06)
 _/ |\__'_|_|_|\__'_|  |  Official https://julialang.org/ release
|__/                   |

(myproject) pkg> st
      Status `~/myproject/Project.toml`
→ [052768ef] CUDA v3.11.0
→ [0c46a032] DifferentialEquations v7.1.0
→ [33e6dc65] MKL v0.5.0
→ [da04e1cc] MPI v0.19.2
        Info packages marked with → not downloaded, use `instantiat
e` to download

(myproject) pkg> instantiate
```

Julia code can be
fast and scalable.

Example: Good scaling of PDE codes



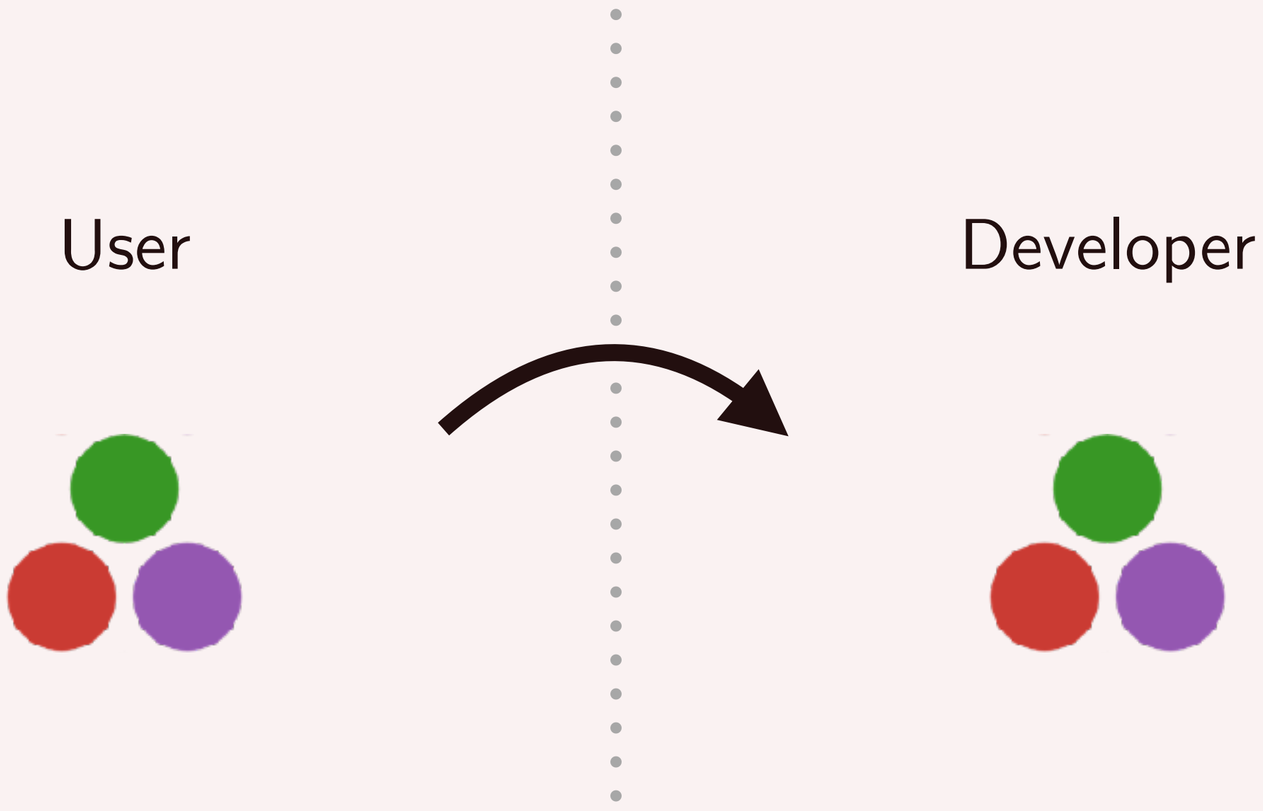
Trixi.jl (Multi-CPU)



ParallelStencil.jl (Multi-GPU)

Julia invites you to gradually delve deeper.

Julia makes it easier to become a developer.



Gradual transition

Let us delve deeper!